

Deep Learning Models for Weed Species Detection from UAV Images



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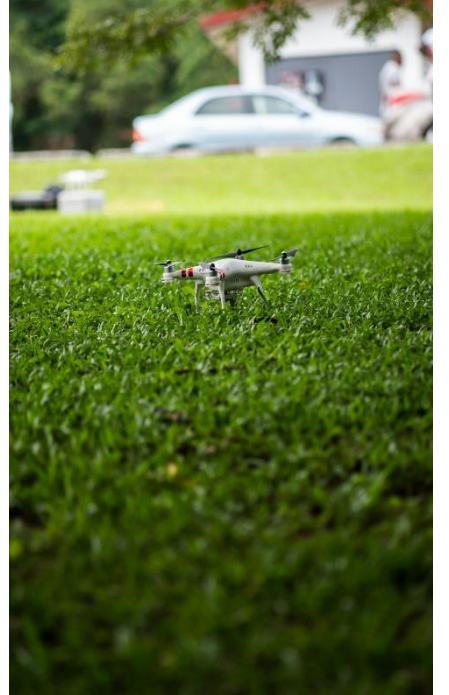
Campus Kongsberg
CS4020-1 Data Science

Introduction:

- Weeds are a major challenge in agriculture because they affect crop growth and productivity
- Traditional weed monitoring is time-consuming and labor-intensive
- UAV-based imaging enables faster and wider observation of agricultural fields
- In this project, UAV images were used to support automatic weed detection
- This approach can contribute to more efficient field monitoring and precision agriculture

Problem and Motivation:

- Weeds compete with crops for water, nutrients, and sunlight
- This can reduce crop yield and increase agricultural costs
- Manual weed detection is slow and difficult, especially in large fields
- UAV images provide a faster and more efficient way to monitor agricultural areas
- Automatic weed detection can support precision agriculture

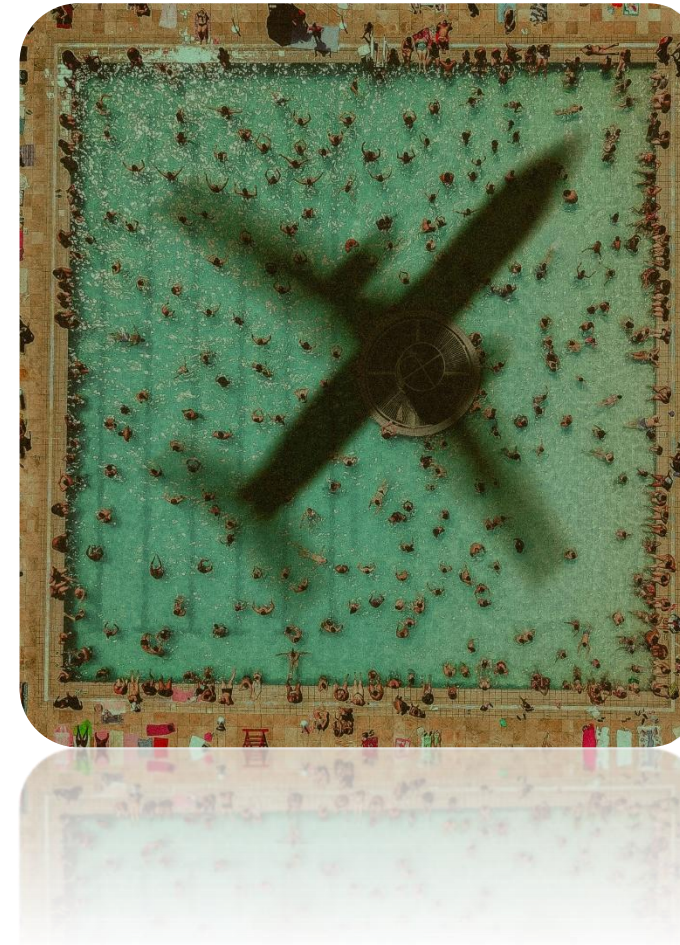


Dataset:

- The CampanetaWeed dataset was used in this project
- We focused on the latest flight: flight_20250411
- The data was collected by a UAV in a fruit orchard in Corbera, Valencia
- Only RGB images were used in this project
- Weed objects were annotated using bounding boxes

Data preparation and Augmentation :

- Image and annotation files were verified to ensure reliable data preparation
- All images were resized to 640*640 pixels and normalized
- The dataset was divided into training, validation, and test sets
- Data augmentation was used to improve model generalization
- Horizontal flipping and brightness/contrast adjustments were applied



Machine Learning Baseline:

- A conventional machine learning baseline was developed as an initial benchmark
- Weed samples were cropped from annotated UAV images
- Handcrafted features such as HOG and RGB color features were extracted
- Two classifiers were tested: SVM and Random Forest
- This baseline was used to compare traditional methods with more advanced approaches

Deep Learning Approach:

- Model: YOLOv8 (object detection)
- Task: Detect weed instances

Why YOLOv8:

- Fast and efficient
- Real-time detection
- Works well with UAV images



Configuration, YOLOv8n/s and Challenges:

- Classes: 6 weed types
- Image size: 416
- Batch size: 8
- Epochs: 10

Variants used:

- YOLOv8n (nano)
 - Faster
 - Lower accuracy
- YOLOv8s (small)
 - Slower
 - Better detection capability

Challenges

- Strong class imbalance
- Complex orchard background
- Sensitivity to training configuration
- Significantly higher compute cost after tuning
- Trade off between speed and accuracy (v8n, v8s)

```
model_n = YOLO("yolov8n.pt")

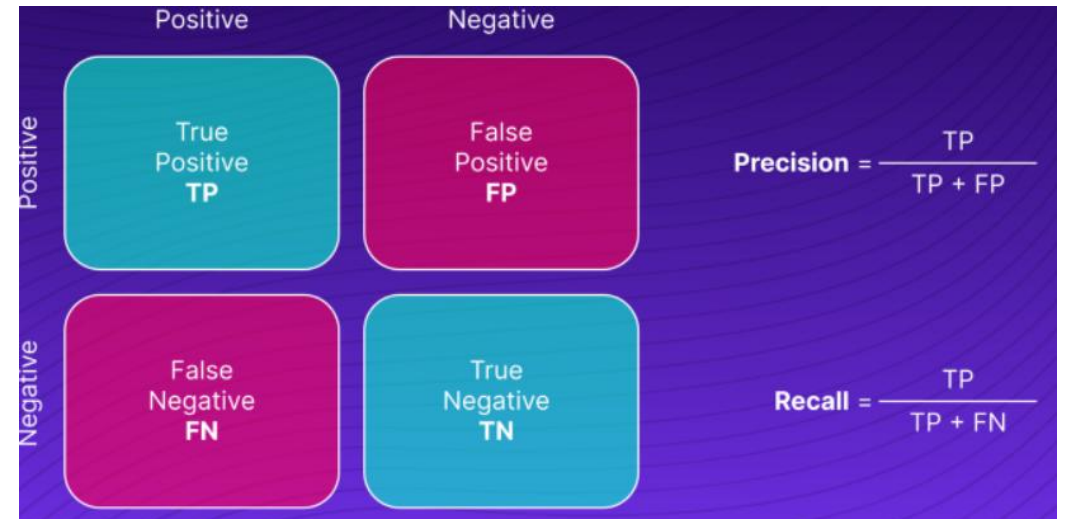
results_n = model_n.train(
    data="data.yaml",
    epochs=50,
    imgsz=640,
    batch=16,
    workers=6,
    cache=True,
    close_mosaic=10,
    augment=True,
    optimizer="SGD",
    project="weed_detection",
    name="yolov8n_model",
    exist_ok=True
)
```

Evaluation Metrics:

- Precision → how many detected weeds are correct
- Recall → how many weeds are successfully detected
- mAP@50 → overall detection accuracy
- mAP@50–95 → stricter evaluation across IoU thresholds

Why these metrics:

- Standard for object detection (YOLO)
- Measure both accuracy and detection coverage



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

How many retrieved items are retrieved?

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Results:

With Initial configuration (epochs=10, imgsz=416, batch=8):

- YOLOv8n:
 - Precision: 0.56
 - Recall: 0.04
 - mAP@50: 0.03
- YOLOv8s:
 - Precision: 0.003
 - Recall: 0.035
 - mAP@50: 0.002

YOLOv8n outperforms YOLOv8s under limited training conditions

• After hyperparameter tuning (epochs=50, imgsz=640, batch=16):

- YOLOv8n:
 - Precision: 0.105
 - Recall: 0.118
 - mAP@50: 0.082
- YOLOv8s:
 - Precision: 0.041
 - Recall: 0.135
 - mAP@50: 0.041

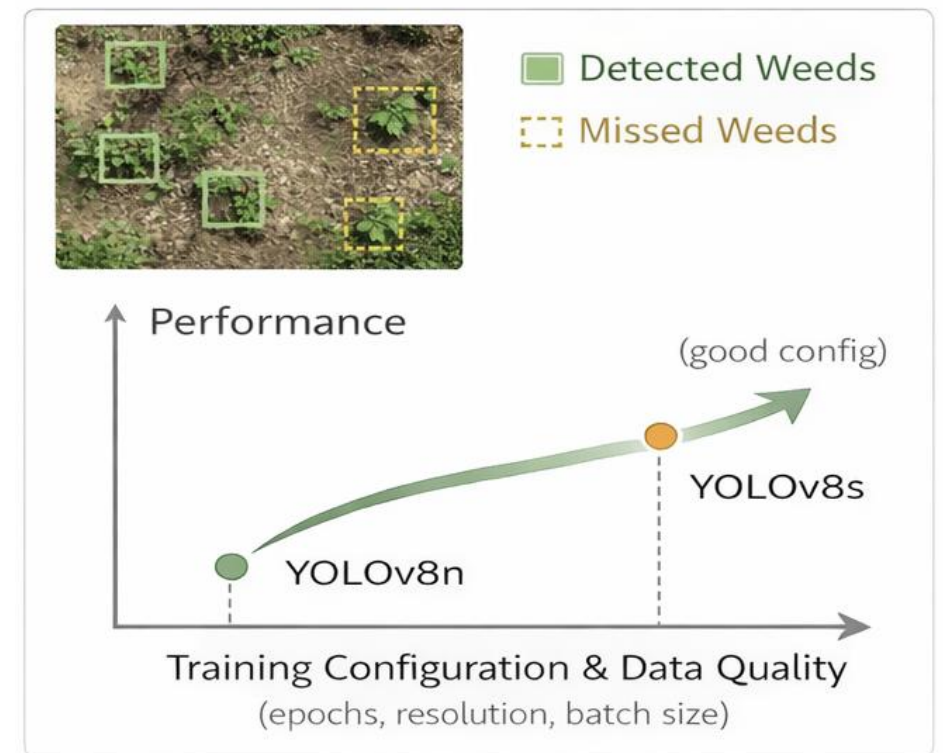
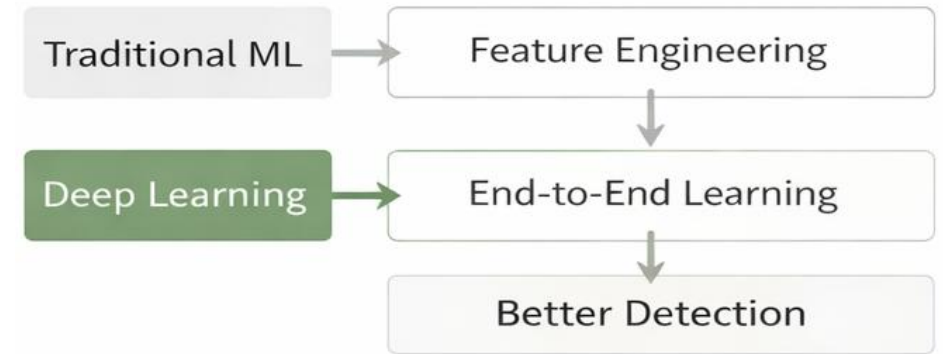
- Both models improve significantly
- YOLOv8s shows stronger improvement in recall

Discussion:

- Deep learning is more suitable for object detection tasks than traditional ML
- Low recall before and after tuning indicates undetected weeds remain untreated
- Model performance depends heavily on:
 - Training configuration (epochs, resolution, batch size, etc..)
 - Data quality

Key insight:

→ Performance is mainly driven by training configuration, especially for larger models



Conclusion:

This project developed a complete pipeline for weed detection using UAV imagery, combining data preparation, preprocessing, machine learning, and deep learning approaches. The results show that deep learning models are more effective than traditional methods for this task. While YOLOv8n performed well under limited training conditions, YOLOv8s achieved better accuracy after proper tuning. Overall, performance was highly dependent on training configuration, highlighting a key trade-off between detection accuracy and computational cost.

Thank You
Questions?